Matrix-Free Second-Order Orbital Optimization for Valence Bond Self-Consistent Field Theory

Tao Xia, Suxin Yu, Wei Wu, Chen Zhou*

State Key Laboratory of Physical Chemistry of Solid Surfaces, Fujian Provincial Key Laboratory of Theoretical and Computational Chemistry, and College of Chemistry and Chemical Engineering, Xiamen University, Xiamen, Fujian 361005

*Email: czhou1989@xmu.edu.cn

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ABSTRACT

Orbital optimization in valence bond self-consistent field (VBSCF) theory must accommodate nonorthogonal orbitals, prescribed atomic orbital (AO) supports, and relaxation of the structure coefficients. We formulate a matrix-free norm-extended optimization (NEO) treatment in support-preserving nonredundant coordinates, retaining orbital–structure coupling through independently expanded orbital and structure response spaces. The numerical study will compare NEO with conventional and block-preconditioned limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimization, with truncated Newton as an additional second-order comparison, reporting convergence, operator work, wall time, and peak memory under matched conditions. [Replace this sentence with the principal quantitative findings and the supported conclusion after the benchmark campaign.]

1 INTRODUCTION

1.1 The optimization problem

Introduce valence bond (VB) theory and valence bond self-consistent field (VBSCF) orbital optimization. Explain the use of nonorthogonal localized orbitals and fixed atomic orbital (AO) supports for hybrid atomic orbitals (HAOs). Separate orbital optimization from the structure-coefficient problem. Introduce overlap-enhanced orbitals (OEOs) as full-AO orbitals, not as an orthogonal-orbital model.

1.2 Relation to second-order multiconfigurational optimization

Review second-order multiconfigurational self-consistent field (MCSCF) methods: restricted-step Newton, augmented-Hessian methods, and norm-extended optimization (NEO), including matrix-free complete active space self-consistent field (CASSCF) implementations [1–3]. Identify what carries over to VBSCF—the local coupled quadratic model and iterative operator actions—and what does not carry over without derivation: orthogonal orbital rotations and their redundancy rules. Introduce limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) optimization as the quasi-Newton comparison.

1.3 Scope of this work

State the methodological contribution as a second-order optimization framework for the specified nonorthogonal VB variational space. Treat integral representation and string contraction as supporting implementations, not independent novelty claims in this paper. The principal numerical question is when additional curvature information improves convergence enough to repay its cost relative to block-preconditioned L-BFGS (block-LBFGS).

2 THEORY

Throughout this section, I, J label VB structures, μ, ν label AO basis functions, i, j label orbitals, n, m label electronic states, and k labels accepted optimization iterations. Boldface denotes matrices and vectors. Calligraphic $\mathcal{H}$ denotes a matrix-free curvature or coupling operator, and square brackets denote its action on a direction. The superscripts o and s identify orbital and structure-coefficient coordinates: the first labels the output space and the second the input space. These labels describe operator actions, not assembled Hessian matrices.

2.1 Nonorthogonal VB wave functions and the stationary structure problem

A valence bond (VB) wave function is expanded in a prescribed set of spin-adapted structures,

$$|\Psi_n(\mathbf{x})\rangle = \sum_I C_{In} |\Phi_I(\mathbf{x})\rangle. \tag{1}$$

where $|\Phi_I\rangle$ is VB structure I, C_{In} is its coefficient in electronic state n, and the sum runs over the selected VB structures. The vector $\mathbf{x}$ contains the allowed atomic orbital (AO) coefficients of the inactive and active orbitals. The Hamiltonian and overlap matrices in the VB structure basis are defined by

$$H_{IJ}(\mathbf{x}) = \langle \Phi_I(\mathbf{x}) | \hat{H} | \Phi_J(\mathbf{x}) \rangle, \qquad M_{IJ}(\mathbf{x}) = \langle \Phi_I(\mathbf{x}) | \Phi_J(\mathbf{x}) \rangle. \tag{2}$$

where $\hat{H}$ is the electronic Hamiltonian. We use $\mathbf{M}$ for the structure overlap matrix and $\mathbf{S}$ for the AO overlap matrix, with elements $S_{\mu\nu} = \langle \chi_\mu | \chi_\nu \rangle$, where χ_μ denotes an AO basis function. The Hamiltonian matrix includes both inactive- and active-electron contributions. The nuclear repulsion is constant during orbital optimization and is omitted from the following expressions.

At fixed orbitals, variation of the structure coefficients gives

$$\mathbf{H}\mathbf{c}_n = E_n\mathbf{M}\mathbf{c}_n, \qquad \mathbf{c}_m^T\mathbf{M}\mathbf{c}_n = \delta_{mn}, \qquad \mathcal{E}(\mathbf{x}) = \frac{1}{N_{\text{state}}}\sum_{n=1}^{N_{\text{state}}} E_n(\mathbf{x}). \tag{3}$$

where E_n and $\mathbf{c}_n$ are the energy and structure-coefficient vector of state n, respectively, with $(\mathbf{c}_n)_I = C_{In}$, and δ_{mn} is the Kronecker delta. We consider real wave functions; the superscript T denotes transpose. The objective $\mathcal{E}$ is the equally weighted average over the lowest N_{state} states, where N_{state} is the number of states included in the average; $N_{\text{state}} = 1$ gives ground-state valence bond self-consistent field (VBSCF) optimization. We assume a positive-definite structure overlap on the working coefficient space and a selected eigenspace separated from excluded roots. Internal degeneracies among the equally weighted states do not change the state-averaged energy. A gap closing between selected and excluded roots, however, can destroy differentiability of this objective.

For an orbital direction, let a dot denote the derivative at the current accepted point. Stationarity with respect to the structure coefficients gives

$$\dot{\mathcal{E}} = \frac{1}{N_{\text{state}}}\sum_{n=1}^{N_{\text{state}}} \mathbf{c}_n^T(\dot{\mathbf{H}} - E_n\dot{\mathbf{M}})\mathbf{c}_n. \tag{4}$$

Here $\dot{\mathbf{H}}$ and $\dot{\mathbf{M}}$ are the directional derivatives of the structure Hamiltonian and overlap matrices, and $\dot{\mathcal{E}}$ is the corresponding energy derivative. Thus no coefficient-response solve is required for the first derivative. Coefficient response is required for the second derivative: the cancellation leading to eq (4) does not permit the eigenvectors to be held fixed when that equation is differentiated.

2.2 Support-preserving orbital coordinates and the update map

The AO expansion of orbital i is

$$\phi_i = \sum_\mu b_{\mu i}\chi_\mu, \qquad \mathbf{b}_i = \mathbf{R}_i\mathbf{x}_i, \tag{5}$$

where $b_{\mu i}$ is an AO coefficient, $\mathbf{b}_i$ is the corresponding full coefficient vector, $\mathbf{x}_i$ contains only its allowed entries, and $\mathbf{R}_i$ embeds them into the full AO basis. These orbital coefficients are distinct from the structure coefficients C_{In}. A hybrid atomic orbital (HAO) has a prescribed localized subset of allowed AOs; an overlap-enhanced orbital (OEO) has full AO support. Neither designation imposes mutual orthogonality of the VB orbitals.

We remove a direction only when it is an infinitesimal invariance of the represented wave function and respects the specified supports. These invariances include orbital scaling and admissible changes of inactive representatives. For a complete active-space expansion with full-AO orbitals, nonsingular changes of the active basis can also be compensated by changes of the structure coefficients. This active-space invariance does not generally hold for an arbitrary selected set of VB structures. Consequently, active–active mixing is not removed merely because both orbitals are active.

The support condition must be imposed on the gauge generators themselves. If $\mathbf{G}_i$ collects candidate gauge generators for orbital i, and $\mathbf{F}_i$ extracts its forbidden AO rows, admissible gauge combinations satisfy

$$\mathbf{F}_i\mathbf{G}_i\mathbf{t} = \mathbf{0}. \tag{6}$$

Here $\mathbf{t}$ contains the coefficients of a linear combination of gauge generators; $\mathbf{F}_i\mathbf{G}_i\mathbf{t} = 0$ requires that combination to vanish on every forbidden AO. The complement of these admissible generators defines a local quotient basis. In particular, truncating a dense gauge generator to the allowed rows is not equivalent to eq (6). Local metric conditioning of the quotient basis produces a lift $\mathbf{U}_k$ from reduced coordinates $\mathbf{p}$ to packed raw coefficients.

The orbital update and the preparation of orbitals for energy evaluation are distinct maps:

$$\mathbf{x}(\mathbf{p}) = \mathbf{x}_k + \mathbf{U}_k\mathbf{p}, \qquad \boldsymbol{\theta}(\mathbf{p}) = \mathcal{P}(\mathbf{x}_k + \mathbf{U}_k\mathbf{p}). \tag{7}$$

where $\mathbf{x}_k$ is the coefficient vector at accepted iteration k, $\mathbf{p}$ is the nonredundant orbital step, and $\mathbf{U}_k$ maps that step to changes in the allowed AO coefficients. The preparation map $\mathcal{P}$ includes orbital normalization and inactive-space projection used by the objective, together with admissible representative choices; $\boldsymbol{\theta}$ denotes the resulting prepared orbital coefficients. The lift is held fixed within a local model and rebuilt after an accepted step. Equation (7) is an additive coefficient update followed by preparation, not an orthogonal orbital rotation or a sphere exponential map.

All derivatives must refer to this same composition. For a scalar function F of prepared orbitals, its Hessian pullback is

$$D^2(F \circ \boldsymbol{\theta})[\mathbf{u}, \mathbf{v}] = D^2F[D\boldsymbol{\theta}\mathbf{u}, D\boldsymbol{\theta}\mathbf{v}] + DF[D^2\boldsymbol{\theta}[\mathbf{u}, \mathbf{v}]]. \tag{8}$$

where D and D^2 denote first and second differentials, $\mathbf{u}$ and $\mathbf{v}$ are reduced orbital directions, and all differentials are evaluated at $\mathbf{p} = 0$. The second term is necessary even though the raw lift in eq (7) is linear. For example, normalization of a raw orbital gives

$$\begin{aligned} \bar{\mathbf{b}} &= \frac{\mathbf{b}}{\|\mathbf{b}\|_\mathbf{S}}, \qquad \|\mathbf{b}\|_\mathbf{S} = (\mathbf{b}^T\mathbf{S}\mathbf{b})^{1/2}, \\ D\bar{\mathbf{b}}[\delta\mathbf{b}] &= \frac{(\mathbf{I} - \bar{\mathbf{b}}\bar{\mathbf{b}}^T\mathbf{S})\delta\mathbf{b}}{\|\mathbf{b}\|_\mathbf{S}}. \end{aligned} \tag{9}$$

where $\mathbf{b}$ is a raw orbital coefficient vector, $\|\mathbf{b}\|_\mathbf{S}$ is its AO-metric norm, $\bar{\mathbf{b}}$ is the normalized vector, $\delta\mathbf{b}$ is a coefficient variation, and $\mathbf{I}$ is the identity matrix of the appropriate dimension. Using an orbital-rotation Hessian with this coefficient update would omit the required map derivatives.

2.3 Physical orbital step metric

The trust-region norm measures changes of physical orbital representatives rather than the Euclidean magnitude of raw AO coefficients. Let $\mathcal{L}_k$ map a packed coefficient tangent to AO-metric-weighted horizontal orbital features. For individually meaningful active orbitals, these features are the differentials of normalized, inactive-projected orbital rays, weighted by $\mathbf{S}^{1/2}$. For inactive orbitals, and for the active orbitals of a complete full-AO active space, horizontal subspace tangents replace individual-ray tangents; inverse Gram factors remove dependence on the chosen basis within each subspace. Variations of the inactive projector are included in the active features.

On the nonredundant coordinates, the metric and its dual norm are

$$\mathbf{G}_o = \mathbf{U}_k^T\mathcal{L}_k^T\mathcal{L}_k\mathbf{U}_k, \qquad \|\mathbf{p}\|^2_{\mathbf{G}_o} = \mathbf{p}^T\mathbf{G}_o\mathbf{p}, \qquad \|\mathbf{r}_o\|^2_{\mathbf{G}_o^{-1}} = \mathbf{r}_o^T\mathbf{G}_o^{-1}\mathbf{r}_o. \tag{10}$$

where $\mathbf{G}_o$ is the physical orbital metric and $\mathbf{r}_o$ is an orbital gradient or stationarity residual. The subscript o denotes orbital coordinates. We assume $\mathbf{G}_o$ is positive definite on the retained coordinates. Metric application and dual-norm evaluation need not construct a dense matrix. Local conditioning of each orbital quotient basis does not, in general, whiten the complete coupled physical metric. The metric is also distinct from a Hessian preconditioner: the former defines step size and residual norms, whereas the latter accelerates the iterative solution without changing the target equation.

2.4 Horizontal structure response and the coupled quadratic model

Collect the selected eigenvectors in $\mathbf{C} = (\mathbf{c}_1, \ldots, \mathbf{c}_{N_{\text{state}}})$. Differentiation of $\mathbf{C}^T\mathbf{M}\mathbf{C} = \mathbf{I}$ fixes the symmetric selected-space component of their response. We choose the internal gauge

$$\dot{\mathbf{C}}_g = -\tfrac{1}{2}\mathbf{C}(\mathbf{C}^T\dot{\mathbf{M}}\mathbf{C}), \qquad \dot{\mathbf{c}}_n = \dot{\mathbf{c}}_{g,n} + \frac{\mathbf{Z}\mathbf{q}_n}{d_n}, \quad d_n = \sqrt{2/N_{\text{state}}}. \tag{11}$$

Here $\dot{\mathbf{C}}_g$ is the selected-space response required by normalization, $\dot{\mathbf{c}}_{g,n}$ is its column for state n, and $\mathbf{q}_n$ contains the independent response coordinates for that state. The factor d_n accounts for the equal state weight. The matrix $\mathbf{Z}$ spans the complement of the selected states and satisfies $\mathbf{C}^T\mathbf{M}\mathbf{Z} = 0$ and $\mathbf{Z}^T\mathbf{Z} = \mathbf{I}$. The latter condition fixes the response coordinates; it does not impose orthogonality on the VB structures. Rotations within the selected equally weighted eigenspace leave the averaged energy unchanged and are excluded from the combined response vector $\mathbf{q} = (\mathbf{q}_1^T, \ldots, \mathbf{q}_{N_{\text{state}}}^T)^T$.

Writing $\dot{\mathbf{H}}[\mathbf{p}]$ and $\dot{\mathbf{M}}[\mathbf{p}]$ for orbital directional derivatives, differentiation of the generalized eigenproblem and projection with $\mathbf{Z}^T$ give

$$\begin{aligned}(\mathcal{H}^{ss}[\mathbf{q}])_n &= \mathbf{Z}^T(\mathbf{H} - E_n\mathbf{M})\mathbf{Z}\mathbf{q}_n,\\ (\mathcal{H}^{so}[\mathbf{p}])_n &= d_n\mathbf{Z}^T\left[(\dot{\mathbf{H}}[\mathbf{p}] - E_n\dot{\mathbf{M}}[\mathbf{p}])\mathbf{c}_n + (\mathbf{H} - E_n\mathbf{M})\dot{\mathbf{c}}_{g,n}[\mathbf{p}]\right].\end{aligned} \tag{12}$$

where $\mathcal{H}^{ss}$ is the structure curvature operator and $\mathcal{H}^{so}$ maps an orbital direction to a structure-response forcing. The subscript n extracts the response for state n. The first action uses structure Hamiltonian and overlap actions on $\mathbf{Z}\mathbf{q}_n$; it does not require assembling a structure Hessian. The derivative of the eigenvalue drops out because $\mathbf{Z}^T\mathbf{M}\mathbf{c}_n = 0$. The factors d_n absorb the state weights and the factor of two in the real-wave-function coefficient Hessian. The gauge term in eq (12) is essential for a moving nonorthogonal structure metric.

Let $\mathcal{H}^{oo}[\mathbf{v}]$ denote the orbital Hessian–vector product (HVP) after enforcing this selected-space normalization gauge and the orbital-map pullback, but before eliminating horizontal structure response. It is not the raw HVP obtained by freezing all structure coefficients. To define its bilinear form, introduce $\mathbf{K}_n = \mathbf{H} - E_n\mathbf{M}$, $\mathbf{K}_{n,u} = D\mathbf{H}[\mathbf{u}] - E_nD\mathbf{M}[\mathbf{u}]$, and $\mathbf{K}_{n,uv} = D^2\mathbf{H}[\mathbf{u},\mathbf{v}] - E_nD^2\mathbf{M}[\mathbf{u},\mathbf{v}]$, with E_n held at its accepted-point value in these definitions. The second variation of the normalization-constrained energy gives

$$\begin{aligned}\mathbf{u}^T\mathcal{H}^{oo}[\mathbf{v}] = \frac{1}{N_{\text{state}}}\sum_n \big\{&\mathbf{c}_n^T\mathbf{K}_{n,uv}\mathbf{c}_n + 2\dot{\mathbf{c}}_{g,n}[\mathbf{u}]^T\mathbf{K}_{n,v}\mathbf{c}_n\\ &+ 2\dot{\mathbf{c}}_{g,n}[\mathbf{v}]^T\mathbf{K}_{n,u}\mathbf{c}_n + 2\dot{\mathbf{c}}_{g,n}[\mathbf{u}]^T\mathbf{K}_n\dot{\mathbf{c}}_{g,n}[\mathbf{v}]\big\}.\end{aligned} \tag{13}$$

Here $\mathbf{u}^T\mathcal{H}^{oo}[\mathbf{v}]$ is the bilinear second variation in orbital directions $\mathbf{u}$ and $\mathbf{v}$. This equation defines the curvature action mathematically; it does not prescribe constructing the directional derivative

matrices or a Hessian. The sum runs over the N_{state} selected states. All matrix derivatives in this expression include the orbital preparation map. Second derivatives of the coefficient gauge cancel against the normalization multipliers at a stationary structure solution; the first derivatives of that gauge remain. The constrained energy expansion is

$$Q_k(\mathbf{p},\mathbf{q}) = \mathcal{E}_k + \mathbf{g}_o^T\mathbf{p} + \tfrac{1}{2}\mathbf{p}^T\mathcal{H}^{oo}[\mathbf{p}] + \mathbf{q}^T\mathcal{H}^{so}[\mathbf{p}] + \tfrac{1}{2}\mathbf{q}^T\mathcal{H}^{ss}[\mathbf{q}], \tag{14}$$

where Q_k is the local quadratic energy model, $\mathcal{E}_k = \mathcal{E}(\mathbf{x}_k)$ is the accepted-point energy, and $\mathbf{g}_o$ is its reduced orbital gradient, obtained from eq (4) in the coordinates of eq (7). The reverse coupling action is denoted by $\mathcal{H}^{os}$, with $\mathbf{p}^T\mathcal{H}^{os}[\mathbf{q}] = \mathbf{q}^T\mathcal{H}^{so}[\mathbf{p}]$. All curvature and coupling actions therefore derive from the same constrained energy, without requiring explicit Hessian assembly.

For a separated lowest-state manifold, $\mathcal{H}^{ss}$ is positive definite on the horizontal space. Define the response map $\mathcal{T}[\mathbf{f}]$ as the solution of $\mathcal{H}^{ss}[\mathcal{T}[\mathbf{f}]] = \mathbf{f}$. If this operator is singular, the minimum-norm solution is defined only for admissible forcing in its range. Eliminating structure response yields

$$\begin{aligned} \mathbf{q}(\mathbf{p}) &= -\mathcal{T}[\mathcal{H}^{so}[\mathbf{p}]], \\ \mathcal{H}_{\text{rel}}[\mathbf{p}] &= \mathcal{H}^{oo}[\mathbf{p}] - \mathcal{H}^{os}[\mathcal{T}[\mathcal{H}^{so}[\mathbf{p}]]]. \end{aligned} \tag{15}$$

where $\mathcal{H}_{\text{rel}}[\mathbf{p}]$ is the relaxed orbital HVP, including structure-coefficient relaxation. The response map is applied by solving a linear equation, not by constructing an inverse. Small structure gaps can make the relaxation contribution large even when the coupling action is small. A complete eigenvalue spectrum is not needed to evaluate eq (15); iterative actions on the admissible response space suffice.

2.5 Orbital restricted steps and the norm-extended formulation

We restrict the physical orbital displacement, not the joint length of orbital and structure response:

$$\min_{\mathbf{p},\mathbf{q}} Q_k(\mathbf{p},\mathbf{q}) \quad \text{subject to} \quad \mathbf{p}^T\mathbf{G}_o\mathbf{p} \le \Delta_k^2. \tag{16}$$

where $\Delta_k > 0$ is the trust-region radius at accepted iteration k. The structure variables describe local electronic relaxation and do not receive an independent trust-region penalty. The stationary equations are

$$\begin{aligned} \mathcal{H}^{oo}[\mathbf{p}] + \mathcal{H}^{os}[\mathbf{q}] + \lambda\mathbf{G}_o\mathbf{p} &= -\mathbf{g}_o, \\ \mathcal{H}^{so}[\mathbf{p}] + \mathcal{H}^{ss}[\mathbf{q}] &= \mathbf{0}, \\ \lambda \ge 0, \qquad \lambda(\mathbf{p}^T\mathbf{G}_o\mathbf{p} - \Delta_k^2) &= 0. \end{aligned} \tag{17}$$

where λ is the nonnegative Lagrange multiplier for the orbital norm constraint. The last equality is the complementarity condition: λ can be positive only for a boundary step. Under the assumptions of eq (15), elimination gives $\mathcal{H}_{\text{rel}}[\mathbf{p}] + \lambda\mathbf{G}_o\mathbf{p} = -\mathbf{g}_o$. The exact minimum additionally requires positive semidefiniteness of the shifted relaxed Hessian. A small stationarity residual alone does not certify that curvature condition.

Norm-extended optimization (NEO) connects a restricted Newton step to a gradient-scaled augmented eigenproblem [1]. In the eliminated orbital space, this relation reads

$$\begin{aligned} \alpha\mathbf{g}_o^T\mathbf{y} &= \mu\beta, \\ \alpha\beta\mathbf{g}_o + \mathcal{H}_{\text{rel}}[\mathbf{y}] &= \mu\mathbf{G}_o\mathbf{y}. \end{aligned} \tag{18}$$

where α scales the gradient coupling, μ is an augmented generalized eigenvalue, and $(\beta, \mathbf{y}^T)^T$ is its eigenvector, with scalar homogeneous component β and orbital component $\mathbf{y}$. For $\alpha > 0$ and $\beta \neq 0$, the extracted step satisfies

$$\mathbf{p} = \frac{\mathbf{y}}{\alpha\beta}, \qquad \lambda = -\mu, \qquad \mathcal{H}_{\mathrm{rel}}[\mathbf{p}] + \lambda \mathbf{G}_o \mathbf{p} = -\mathbf{g}_o, \qquad \alpha^2 = \frac{\lambda}{-\mathbf{g}_o^T \mathbf{p}}. \tag{19}$$

Here the augmented eigenvalue determines the same orbital shift λ that appears in eq (17). The gradient scale is therefore tied to the restricted step rather than being an arbitrary fixed parameter. The unshifted interior Newton step is recovered as a limiting case; a zero homogeneous component or a hard case requires direct treatment of the restricted quadratic problem rather than division by β. Our solver obtains the small projected restricted step spectrally and uses eqs (18) and (19) to express its norm-extended interpretation. It does not repeatedly solve a full-dimensional augmented eigenproblem or form $\mathcal{H}_{\mathrm{rel}}$.

2.6 Independent response spaces and full residual certification

Orbital and structure spaces are expanded independently. Let $\mathbf{V}$ and $\mathbf{W}$ contain the retained orbital and structure basis directions, respectively. We write the trial response as $\mathbf{p} = \mathbf{V}\boldsymbol{\xi}$ and $\mathbf{q} = \mathbf{W}\boldsymbol{\zeta}$, where $\boldsymbol{\xi}$ and $\boldsymbol{\zeta}$ are the corresponding expansion coefficients. The projected matrices are

$$\begin{aligned} \mathbf{h}^{oo} &= \mathbf{V}^T \mathcal{H}^{oo}[\mathbf{V}], & \mathbf{h}^{so} &= \mathbf{W}^T \mathcal{H}^{so}[\mathbf{V}], \\ \mathbf{h}^{ss} &= \mathbf{W}^T \mathcal{H}^{ss}[\mathbf{W}], & \mathbf{G}_V &= \mathbf{V}^T \mathbf{G}_o \mathbf{V}, \\ \mathbf{h}_{\mathrm{rel}} &= \mathbf{h}^{oo} - \mathbf{h}^{os}(\mathbf{h}^{ss})^{\dagger}\mathbf{h}^{so}. \end{aligned} \tag{20}$$

where $\mathbf{h}^{oo}$, $\mathbf{h}^{so}$, $\mathbf{h}^{ss}$, and $\mathbf{G}_V$ are small subspace matrices, $\mathbf{h}^{os} = (\mathbf{h}^{so})^T$, and $\mathbf{h}_{\mathrm{rel}}$ is the projected relaxed curvature matrix. An operator acting on a basis matrix applies independently to its columns. Only these subspace projections are stored as matrices; the full curvature and coupling operators remain matrix-free. The dagger on the small matrix denotes its Moore–Penrose pseudoinverse. New directions require new operator images; previously computed images are retained at the same accepted orbital point. Changing the trust radius does not invalidate these images. The projected Schur complement is inexpensive once these small matrices are available, but is not assumed to reproduce the full structure relaxation merely because its projected equations are solved accurately.

Certification uses residuals in the original retained coordinate spaces:

$$\begin{aligned} \mathbf{r}_o &= \mathbf{g}_o + \mathcal{H}^{oo}[\mathbf{p}] + \mathcal{H}^{os}[\mathbf{q}] + \lambda \mathbf{G}_o \mathbf{p}, \\ \mathbf{r}_s &= \mathcal{H}^{so}[\mathbf{p}] + \mathcal{H}^{ss}[\mathbf{q}], \\ \left(\|\mathbf{r}_o\|^2_{\mathbf{G}_o^{-1}} + \|\mathbf{r}_s\|_2^2\right)^{1/2} &\leq \epsilon_{\mathrm{abs}} + \eta_k \|\mathbf{g}_o\|_{\mathbf{G}_o^{-1}}. \end{aligned} \tag{21}$$

where $\mathbf{r}_o$ and $\mathbf{r}_s$ are the orbital and structure stationarity residuals, respectively, $\epsilon_{\mathrm{abs}} \geq 0$ is an absolute residual tolerance, and $\eta_k \geq 0$ is the iteration-dependent relative forcing tolerance. The notation $\|\cdot\|_2$ denotes the Euclidean norm; for the structure residual it refers specifically to the scaled coordinates in eq (11). Step feasibility and complementarity are checked separately. Curvature information from the projected space is not a general certificate of positive semidefiniteness outside that space.

A small structure residual can still give a significant orbital error when $\mathcal{H}^{ss}$ is ill-conditioned. Where needed, we solve for a response correction and examine its orbital effect:

$$\mathcal{H}^{ss}[\delta\mathbf{q}] = -\mathbf{r}_s, \qquad \mathbf{r}_o^{\mathrm{corr}} = \mathbf{r}_o + \mathcal{H}^{os}[\delta\mathbf{q}]. \tag{22}$$

Here $\delta\mathbf{q}$ is a correction to the candidate structure response and $\mathbf{r}_o^{\mathrm{corr}}$ is the orbital residual after that correction. An unresolved correction supplies an additional response direction; a sufficiently accurate correction completes the candidate response. This targets the response needed by the current orbital candidate instead of solving every orbital basis direction to final response accuracy. It does not eliminate coupling work: appending an orbital direction still requires its orbital-to-structure image.

2.7 Trial evaluation, approximate models, and local convergence

Only the orbital component of the local step is applied in eq (7). At the trial orbitals, the selected structure problem is solved again; $\mathbf{q}$ is an auxiliary local response, not a replacement for this nonlinear reoptimization. Acceptance compares the actual relaxed energy decrease with the coupled-model prediction,

$$\rho_k = \frac{\mathcal{E}_k - \mathcal{E}(\mathbf{x}_k + \mathbf{U}_k\mathbf{p})}{Q_k(\mathbf{0},\mathbf{0}) - Q_k(\mathbf{p},\mathbf{q})}. \tag{23}$$

where ρ_k is the ratio of actual to predicted energy decrease. The numerator uses the reoptimized structure energy at the trial orbitals, whereas the denominator uses the local quadratic model. Accepted points receive a new true gradient, chart, and local model. Rejected trials retain the accepted-point operator data while the trust radius is adjusted. No cross-iteration secant Hessian history is required by this NEO formulation; a previous orbital direction may provide a warm start.

The full coupled model must be distinguished from the production core-only approximation. The latter retains direct inactive/core response and the nonlinear pullback with fixed upstream adjoints, but omits both local active response and horizontal structure response. It is therefore not simply eq (14) with $\mathcal{H}^{so} = 0$, nor is it the complete fixed-structure orbital Hessian. The current admission decision compares response-space dimension and estimated contraction work with orbital-space dimension and core work. This is a computational work heuristic, not an error bound on omitted curvature. In both branches, trial energies and accepted gradients are evaluated for the same relaxed VBSCF objective. Full and core-only calculations must be identified separately in the numerical results.

Near a smooth isolated minimum with a nonsingular quotient Hessian, the full model approaches an ordinary Newton step once the trust region is inactive. Superlinear local convergence requires the effective Newton residual to be $o(\|\mathbf{g}_o\|)$; quadratic convergence requires an $O(\|\mathbf{g}_o\|^2)$ residual, locally Lipschitz second derivatives, and correspondingly controlled structure-solver and derivative errors. A forcing sequence $\eta_k = O(\|\mathbf{g}_o\|)$ supplies the relative part of this requirement, but a fixed nonzero ϵ_{abs} eventually limits the certified accuracy. These local conclusions do not automatically apply to the core-only model or establish rapid entry into the Newton neighborhood.

One-electron orbital blocks are used as a preconditioner, not substituted for the full Hessian in these statements. The comparison methods are block-preconditioned L-BFGS (block-LBFGS) and truncated Newton with Hessian–vector products (TNHVP), where a Hessian–vector product (HVP) applies the Hessian to a direction without forming the matrix. Block-LBFGS retains secant history

with a block initial inverse, whereas the current TNHVP implementation uses a block-LBFGS baseline with conditional HVP-based correction. Their performance must be measured against NEO rather than inferred from the labels "first order" and "second order."

3 IMPLEMENTATION

3.1 Matrix-free derivative actions

Describe the accepted-point data required for gradient and HVP evaluation: normalization, inactive projection, integral transformation, structure action, and response. Identify reused quantities and directional quantities, without claiming that matrix-free means no large intermediates. Give memory costs in orbital, structure, and unique-string dimensions.

3.2 Trial steps and accepted-point updates

Give the acceptance and trust-radius update settings associated with eq (23). State which quantities are refreshed at an accepted point and which are reused during subspace expansion. Count rejected trials as work even when they do not advance the accepted-iteration counter.

Algorithm 1. Accepted-point second-order VBSCF optimization (outline)

1. Read the same orbital guess and structure list used by all comparison methods.
2. Solve the selected structure problem; evaluate energy and the physical orbital gradient.
3. Build the support-preserving local coordinates, metric, derivative actions, and preconditioner.
4. Expand the iterative second-order model; certify its residual and record response treatment.
5. Retract the orbital step, solve the trial structure problem, and evaluate model agreement.
6. Accept or reject the trial; refresh accepted-point data and test both stopping criteria.

3.3 Cost accounting

Report orbital HVPs, orbital-to-structure and transpose-coupling actions, structure Hamiltonian/overlap actions, trial evaluations, and accepted steps separately. Include setup, final observables, and total process memory in end-to-end measurements; isolate optimizer time in an additional breakdown.

4 COMPUTATIONAL DETAILS

4.1 Methods and matched conditions

Method	Selector	Role
XMVB-style L-BFGS	ISCF=5	Conventional quasi-Newton baseline

Method	Selector	Role
TNHVP	ISCF=7	Matrix-free truncated Newton comparison
Block-LBFGS	ISCF=8	Enhanced baseline with one-electron block initial inverse
NEO	ISCF=9	Principal norm-extended second-order method

Table 1. Comparison methods. Input selectors are implementation details, not method names.

Use identical initial orbitals, geometry, basis, active electrons/orbitals, spin, selected structure list, state weights, integral representation, and final convergence tests. Compare exact integrals and the resolution-of-the-identity (RI) approximation only in separate matched groups. Dense and Davidson structure solvers must solve the same selected generalized eigenproblem to controlled accuracy.

Record the projected-gradient convention and the physical dual-metric norm. The working energy-change and projected-gradient targets are 10^{-7} hartree and 10^{-3}, respectively; confirm the precise implemented norms and final settings before collection. Use an independently tighter calculation for endpoint comparisons.

4.2 Test-set design

Group	Examples	Purpose
Small derivative oracles	F_2; small selected-structure cases	Finite differences, dense Hessian, endpoint checks
Localized HAO	241 (benzene), 240, 7975, 7963, YAMSAI	Support constraints and convergence variation
Transition metals	MnF_2 and $FeCl_2$	Conditioning, spin, and OEO/HAO behavior
Larger response spaces	CERRAS and LOFLEA	Response cost and memory
Equal-weight state average	Small multistate cases; selected pyrrole/OEO cases	Root-space and response stability

Table 2. Proposed test groups; molecular specifications and eligibility remain to be frozen.

Do not include unresolved endpoint discrepancies as successful validation. A PySCF CASSCF comparison is a variational-space reference only when the OEO structure expansion, spin/symmetry sector, integrals, and core convention match; otherwise it is a diagnostic comparison.

4.3 Hardware and reproducibility

Collect publication timings on Hanhai25 compute nodes through Slurm. Report the central processing unit (CPU) model, allocated cores, affinity, OpenMP and Basic Linear Algebra Subprograms

(BLAS) thread counts, compiler flags, dependency versions, and commit identifier. Record peak resident set size (RSS) at the job/process level. Preserve nonconverged runs and resource failures in the dataset rather than dropping them.

5 RESULTS AND DISCUSSION

5.1 Derivative and response validation

Table 3 will report gradient directional differences, HVP directional differences, bilinear symmetry defects, and coupled-versus-Schur agreement on small cases. Include nonorthogonal well-conditioned and near-singular orbital points, HAO and OEO, and at least one equal-weight state average. [Insert verified numerical errors.]

5.2 Local convergence

Figure 2 will show energy error and physical gradient norm for identical controlled perturbations around a common stationary point. Establish the local convergence regime separately from the number of iterations required to enter it. [Insert trajectories and the residual forcing sequence.]

5.3 End-to-end comparison

Figures 3 and 4 will plot accepted steps and total wall time against a common system ordering, with failures visibly marked. Table 4 will provide final energies, energy changes, physical gradients, action counts, wall time, and RSS. Discuss cases where fewer steps do not yield lower wall time. [Insert matched results; do not mix revisions.]

5.4 Where the second-order cost is spent

Figure 5 will break down setup, core HVP, coupling/structure response, trial evaluation, and final output. Analyze larger active spaces using measured work and memory rather than accepted-step count alone. Compare full and restricted response with the model treatment stated for every run. [Insert stage timings and response statistics.]

5.5 Ablations and limitations

Separate model changes from preconditioner changes. Prioritize matched full/restricted response and one-electron block/Hessian-diagonal comparisons; identity preconditioning need not be a production benchmark. Discuss small structure gaps, incomplete selected-VB spaces, root changes, and redundant-coordinate conditioning. [Insert only ablations that answer a specific algorithmic question.]

6 CONCLUSIONS

Summarize the validated second-order formulation, the measured range of efficiency, and the cost of structure relaxation. State which method is recommended for which workload only if the data support that distinction. [Write the quantitative conclusion after Sections 5.1–5.5 are complete.]

ASSOCIATED CONTENT

Supporting Information

Planned contents of the Supporting Information (SI): coordinate and response derivations; update-map derivatives; finite-difference protocols; complete molecular specifications and initial orbitals; raw convergence trajectories; nonconvergence records; timing scripts and environment details. Submit the SI separately from the main manuscript.

AUTHOR INFORMATION

Corresponding Author: Chen Zhou. Email: czhou1989@xmu.edu.cn. Author Contributions: [To be agreed by all authors]. Notes: [Competing-interest statement to be confirmed].

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[Funding, computing resources, and individual acknowledgments to be confirmed. Add any required AI-assistance disclosure after review of the applicable ACS policy; do not list an AI tool as an author.]

DATA AVAILABILITY STATEMENT

[Supply a persistent public repository link or DOI for the input files, initial orbitals, raw outputs, benchmark metadata, and analysis scripts. Identify the code release and commit used for every figure. Explain any justified restrictions.]

REFERENCES — STARTER LIST

(1) Jensen, H.-J. Aa.; Jørgensen, P. A Direct Approach to Second-Order MCSCF Calculations Using a Norm Extended Optimization Scheme. *J. Chem. Phys.* **1984**, *80* (3), 1204–1214.

(2) Werner, H.-J.; Knowles, P. J. A Second Order Multiconfiguration SCF Procedure with Optimum Convergence. *J. Chem. Phys.* **1985**, *82* (11), 5053–5063.

(3) Sun, Q.; Yang, J.; Chan, G. K.-L. A General Second Order Complete Active Space Self-Consistent-Field Solver for Large-Scale Systems. *Chem. Phys. Lett.* **2017**, *683*, 291–299.

[Complete the VBSCF/XMVB, inexact-Newton, trust-region, L-BFGS, and density-fitting literature before submission. This starter list is not an exhaustive priority review.]